

Question 1: What is a null hypothesis (H_0) and why is it important in hypothesis testing?

Ans:-

Null Hypothesis (H_0):

The null hypothesis is a statement that there is no effect, no difference, or no relationship between variables.

Importance of H_0 in hypothesis testing:

- It provides a baseline or starting point for analysis
- It helps decide whether observed results are due to chance or a real effect
- Statistical tests are designed to test against the null hypothesis
- It enables objective decision-making using data and probability

Question 2: What does the significance level (α) represent in hypothesis testing?

Ans:-

Significance level (α):

The significance level (α) is the maximum probability of committing a Type I error, that is, rejecting a true null hypothesis.

Question 3: Differentiate between Type I and Type II errors.

Ans:-

Feature	Type I Error	Type II Error
Alternative Name	False Positive	False Negative
Reality	Null Hypothesis is True	Null Hypothesis is False
Decision	Reject Null Hypothesis	Fail to Reject Null Hypothesis
Probability	α (Alpha)	β (Beta)
Example	Testing positive for a disease you don't have	Testing negative for a disease you do have

Question 4: Explain the difference between a one-tailed and two-tailed test. Give an example of each.

Ans:-

Basis	One-tailed Test	Two-tailed Test
Direction	Tests effect in one direction only	Tests effect in both directions
Alternative Hypothesis	$\mu > \mu_0$ or $\mu < \mu_0$	$\mu \neq \mu_0$
Rejection Region	In one tail of the distribution	In both tails of the distribution
Significance Level (α)	Entire α in one tail	α divided into two tails
Example	Mean score greater than 60	Mean score different from 60

Question 5: A company claims that the average time to resolve a customer complaint is 10 minutes.

A random sample of 9 complaints gives an average time of 12 minutes and a standard deviation of 3 minutes. At $\alpha = 0.05$, test the claim.

Ans:-

Hypothesis Test (t-test):

- $H_0: \mu = 10$ minutes
- $H_1: \mu \neq 10$ minutes (two-tailed test)
- Sample mean ($\bar{x}$) = 12
- Standard deviation (s) = 3
- Sample size (n) = 9
- $\alpha = 0.05$

Test statistic:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{12 - 10}{3/\sqrt{9}} = \frac{2}{1} = 2$$

- Degrees of freedom (df) = $n - 1 = 8$
- Critical value ($t_{0.025, 8}$) $\approx \pm 2.306$

Decision:

$|t| = 2 < 2.306$, fail to reject H_0 .

Question 6: When should you use a Z-test instead of a t-test?

Ans:-

Z-test should be used when:

- The population standard deviation (σ) is known, and
- The sample size is large ($n \geq 30$), and
- The data is normally distributed (or approximately normal).

Question 7: The productivity of 6 employees was measured before and after a training program

Employee	Before	After
1	50	55
2	60	65
3	58	59
4	55	58
5	62	63
6	56	59

At $\alpha = 0.05$, test if the training improved productivity.

Ans:-

Paired t-test:

- $H_0: \mu_d = 0$ (no improvement)
- $H_1: \mu_d > 0$ (productivity improved)

Differences (After – Before):

5, 5, 1, 3, 1, 3

- Mean difference ($\bar{d}$) = 3
- Standard deviation (sd) = 1.79
- $n = 6$

$$t = \frac{\bar{d}}{sd/\sqrt{n}} = \frac{3}{1.79/\sqrt{6}} \approx 4.10$$

- $df = 5$
- Critical t (0.05, one-tailed) ≈ 2.015

Decision:

4.10 > 2.015, reject H_0 .

Question 8: A company wants to test if product preference is independent of gender.

Gender	Product A	Product B	Total
Male	30	20	50
Female	10	40	50
Total	40	60	100

At $\alpha = 0.05$, test independence

Ans:-

Step 1: Set hypotheses

- H_0 : Product preference is independent of gender
- H_1 : Product preference is not independent of gender

Step 2: Observed frequencies (O)

Gender	Product A	Product B	Total
Male	30	20	50
Female	10	40	50
Total	40	60	100

Step 3: Expected frequencies (E)

Formula: $E_{ij} = \frac{(\text{Row total})(\text{Column total})}{\text{Grand total}}$

- Male, Product A: $E = \frac{50 \cdot 40}{100} = 20$
- Male, Product B: $E = \frac{50 \cdot 60}{100} = 30$
- Female, Product A: $E = \frac{50 \cdot 40}{100} = 20$
- Female, Product B: $E = \frac{50 \cdot 60}{100} = 30$

Gender	Product A	Product B	Total
Male	20	30	50
Female	20	30	50
Total	40	60	100

Step 4: Compute χ^2 statistic

Formula: $\chi^2 = \sum \frac{(O - E)^2}{E}$

- Male, Product A: $(30 - 20)^2 / 20 = 100 / 20 = 5$
- Male, Product B: $(20 - 30)^2 / 30 = 100 / 30 \approx 3.333$

- Female, Product A: $(10-20)^2/20 = 100/20 = 5$
- Female, Product B: $(40-30)^2/30 = 100 / 3 \approx 3.333$

$$\chi^2 = 5 + 3.333 + 5 + 3.333 = 16.666 \approx 16.67$$

Step 5: Degrees of freedom

$$df = (r - 1) (c - 1)$$

$$= (2 - 1) (2 - 1) = 1$$

Step 6: Critical value at $\alpha = 0.05$, $df = 1$

$$X_{0.051}^2 = 3.841$$

Step 7: Decision

$$X_{calculated}^2 = 16.67 > 3.841 = X_{calculated}^2$$

Reject H_0

There is sufficient evidence at the 5% significance level to conclude that product preference depends on gender.